

**ME594: 3D Robotics, HW 3**

**Homework due on 02-02-2025, Topics: Euler Angles / Quaternions / Axis Angle.**

1. **Rotation/Quaternion/Axis-Angle/Euler Angles:** Suppose  $R$  represents a rotation of  $90^\circ$  about  $y_0$  followed by a rotation of  $45^\circ$  about  $z_1$ .

- (a) Compute  $R$ .
- (b) Compute equivalent quaternion for  $R$ .
- (c) Compute equivalent axis/angle for  $R$ .

- **Gradescope:** Submit your typed/handwritten work. Feel free to use Python if needed.
- **Email:** None

2. **Rotation/Quaternion/Axis-Angle/Euler Angles:** Consider the rotation sequence given by the Euler angles:  $\phi = \pi/6$  (about x) followed by  $\theta = \pi/3$  (about y), and finally  $\psi = \pi/8$  (about z).

- (a) Give the quaternion representation for each individual rotation.
- (b) Take the appropriate quaternion products of the individual quaternions (in a above) to compute the net quaternion.
- (c) Convert the above quaternion (in b above) to a rotation matrix.
- (d) Check the above calculation (in c above) using the rotation formula based on Euler angles.

- **Gradescope:** Submit your typed/handwritten work. Feel free to use Python if needed.
- **Email:** None

### 3. Rott's pendulum in MuJoCo

The Rott's double pendulum is another system (besides the standard double pendulum) which demonstrates chaotic behavior. You can watch a video here: <https://youtu.be/dhZxdV2naw8>. Fig. 1 shows model of the Rott's pendulum. Note that the first link (connected to the ground at O) is shown in blue and is L-shaped while the red link is connected to the blue link via a hinge joint at P. Model the pendulum in xml. Model the pendulum in vertical plane in xml. When the file is loaded using *mujoco viewer* it should show chaotic behavior like that shown in the video. The use of a load key to input a start position is acceptable. Assume  $\ell_1 = 2\ell_2 = 1$ ,  $m_1 = m_2 = 1$ . You can use an inertia of 0.01 for both links. Start the Rott's pendulum such that the longer side of the blue link is vertical (i.e.,  $\theta_1 = 1.57$  rad) and the red link is horizontal (i.e.,  $\theta_2 = 1.57$ ). Gravity is  $g = 9.81$ .

#### HINTS:

- UPDATED 1/29/2025:** This video shows how to model a planar three link pendulum <https://youtu.be/MmWV-7fV4bg>.
  - Ensure that the body frames for the blue and red links are at the pin joints O and P, respectively. That way, the location of the hinge or pin joint would be at the body frame axis.
  - Here is my attempt <https://youtube.com/shorts/c0qpvcuX4V0>
- **Gradescope:** None
  - **Email:** Submit the XML file to pranav@uic.edu. Please add "HW3 Rotts Pendulum" to your subject line. If submitting as a group, please copy your group member.

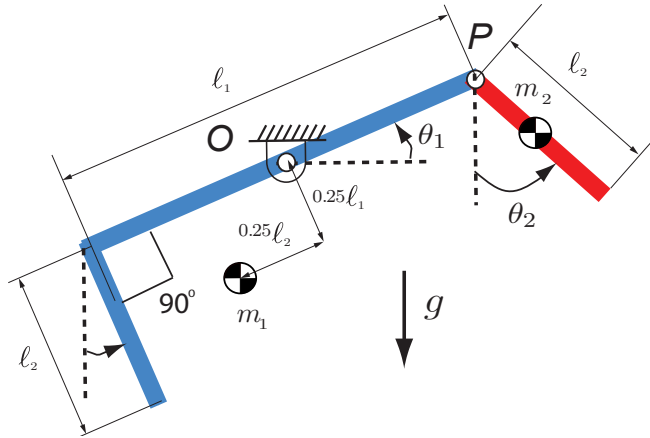


Figure 1: Rott's pendulum